Wildlitics research review and development priorities

Research cutoff: **24 October 2026**. Prepared for Brian Martin.

The next useful development step is to establish a trustworthy local benchmark, improve the training/review data, and connect identification quality to ecological inference. Adding more backbones can then be tested against that baseline. The literature supports several promising methods, but it does not establish Wildlitics' present accuracy or guarantee that any one change will improve it.

Research library

Requested topic	Journal articles in the topic list
1. Enhancing identification models	100
2. Species data analysis and camera-trap workflows	100
3. Detection, accuracy and reducing confusion	100

There are **200 unique journal articles and 300 topic placements**. Papers can appear in more than one topic, especially topics 1 and 3. Every included DOI was checked against publisher-deposited Crossref metadata. This is 100 articles per topic, not 100 different journal titles or 300 different papers.

This is a targeted research review, not a systematic review or a full-text reading of 200 papers. All entries were screened by title and bibliographic metadata; 28 priority sources received closer abstract or accessible-content assessment for the synthesis. Other entries are a reading library, with relevance questions rather than unverified findings. Preprints, conference papers, software documentation and community posts are identified separately and excluded from the journal counts. See METHODOLOGY.md and KEY_EVIDENCE.md.

The searchable library, CSV and RIS bibliography provide the full references. The RIS file can be imported into a reference manager such as Zotero. No paywalled full texts or copied abstracts are bundled.

What the current Wildlitics work already contains

The public README describes version 2.3.3 using MDV6, SpeciesNet 4.0.3a/package 5.0.5 and BioCLIP 2, with human review, camera/deployment metadata, offline maps, descriptive activity reports and optional local BioCLIP-head training. Its public repository does not contain the desktop application's source code.

Draft PR #1 adds experimental annotation exports, grouped dataset splits, timestamped video sampling and a SpeciesNet review importer. Those are separately tested helpers, not an installed-app update. The separate open-method prototype contains retrieval/DINOv2 and document/taxonomy adapters, with real local GLM-OCR/Qwen validation still pending. This research review changes no application models or statistical outputs.

1. Enhancing the identification models

Start with local adaptation of the existing models. A 2026 study comparing global, local and fine-tuned classifiers favoured fine-tuned SpeciesNet on a balanced set of 15 common Australian rainforest species, including withheld-camera tests. That is a strong reason to test the approach locally, with explicit rare-species evaluation. It is not evidence that its reported performance or sample-size plateau transfers to Borneo. Rajmohan et al.

Compare the current Wildlitics model policy with: the existing frozen BioCLIP head; a frozen DINOv2 head; and a modest SpeciesNet fine-tune. Use the same reviewed animal crops, class definitions, held-out sites and test

encounters. Keep reference-photo retrieval as a separately evaluated source of evidence. A correct nearest reference and a correct closed-set classification are different tasks.

Improve label quality and representation of difficult classes. The 2026 training-quality study found that species-level occupancy and activity could suffer for rare or similar taxa even when community richness appeared stable. This argues for label audits, night images, partial views and confusing taxa before an extensive architecture sweep. Bevan et al.

Use human review to acquire useful new data. An active-learning queue should combine uncertainty, model disagreement and coverage of different sites, illumination and rare species. An uncertainty-only queue can keep selecting almost identical frames. Preserve a representative expert-labelled audit sample as well as the actively selected corrections. The camera-trap active-learning literature supports the approach; its reported labelling savings require local verification. Norouzzadeh et al.

Treat augmentation and sequence models as experiments. Background-diverse copy-paste augmentation and attention over related animal crops are promising additions. Apply augmentation only within training partitions, retain source provenance, and test real mixed-species encounters. A sequence model should preserve separate animals and allow multiple species. Copy-paste study, multi-crop attention study

2. Species analysis: useful CameraSweet ideas and stronger statistical methods

Make operational effort and timestamps explicit. CameraSweet's official workflow includes clock correction, camera grouping and matrices for occupancy and abundance-style analyses. Wildlitics should extend its existing deployment records with auditable operating intervals, downtime, time zones, clock edits and processing coverage. Preserve original timestamps as well as corrected values. CameraSweet documentation, Harris et al.

Add compatible analytical exports first. Camtrap DP provides linked deployment, media and observation records. camtrapR supports camera-operation matrices, record tables and analytical workflows. A particularly relevant update was published online on **21 October 2026**: camtrapR version 3's paper describes a dashboard, covariate preparation, activity, occupancy, spatial prediction and model diagnostics. A tested local R adapter or export path is a practical way to build on those methods. Camtrap DP, camtrapR 2026

Distinguish detection history from a presence map. Wildlitics already describes observed presence/nondetection handling. Formal occupancy analysis additionally estimates occurrence and observation processes from repeated occasions. Export **1** for a qualifying detection, **0** for a qualifying operational occasion without a detection, and **NA** for unsampled/inactive occasions or insufficient processing coverage under a documented policy. An observed zero remains a nondetection, with imperfect detection handled by the statistical model. MacKenzie et al.

Give activity filtering its own settings and sensitivity analysis. A 2022 study argues against routine time-to-independence filtering for activity estimates; a 2024-online/2025-issue counter-study finds behavioural situations in which filtering helps. These conflicting results support retaining the raw data and reporting estimator-specific choices. Compare unfiltered, sequence-based and 30/60-minute treatments as sensitivity scenarios where appropriate. Encounter separation, occupancy occasion width and activity sampling units should be stored separately. Peral et al., Smith, occupancy autocorrelation guidelines

Extend descriptive charts with validated estimators. Useful additions are circular temporal overlap with uncertainty, incidence-based diversity/sample-coverage curves, and formal occupancy models with fit diagnostics. A basic activity density curve does not by itself estimate the fraction of time active; that requires additional assumptions. Likewise, camera encounter totals do not establish individual abundance. Ridout and Linkie, Rowcliffe et al., iNEXT

Density estimation should follow later, with a specific supported method and its required survey inputs. Random-encounter approaches require movement and camera-geometry information; capture-recapture requires defensible individual identities. Random encounter model, individual-ID error study

3. Detection, accuracy and species confusion

Measure each stage of the pipeline. Report: animal-box recall and precision against reviewed boxes; conditional species accuracy on valid crops; and end-to-end species recall on all annotated animals, including animals the detector missed. Also record failures, processing time, memory and reviewer effort. Detector confidence, classification scores, reference similarity and ecological detection probability are distinct quantities.

Build a confusion dashboard. Show per-class precision, recall and F1, support in independent encounters, common confusion pairs, unresolved rates and human corrections. Break results down by day/night, site, camera model, apparent animal size and partial/obscured views. Preserve genus/family suggestions and unexpected taxa for review. Avoid forcing a local-checklist match. Habitat and background shift can cause large generalisation losses. Norman et al., Schneider et al.

Calibrate and choose review thresholds using held-out development data. Save full classifier logits when available, assess calibration separately from classification, and choose any per-class acceptance policy using validation/calibration data. Test it once on untouched data. Evaluate confidence reliability and selective risk/coverage, not merely the number of predictions above a threshold. Top-k-only probabilities cannot reconstruct a full-distribution temperature-scaling calculation. The current FAISS cosine scores need their own empirical validation before any probability interpretation. Camera-trap calibration study

Audit both unexpected positives and presumed blanks. Prioritise rare-species positives, model disagreements and difficult night records, while retaining random audits of ordinary and blank predictions. Compare detector variants, threshold changes, context padding or optional slicing on the same holdout; measure their runtime and false-positive tradeoffs. Every downstream classifier depends on the crops it receives.

Validate ecological conclusions independently. Recent studies show both encouraging agreement between AI and expert occupancy models and important failures for rare species or low-precision classifications. The relevant question is whether Wildlitics preserves the ecological result for the intended survey, not whether another study reported high image accuracy. Thornton et al. 2026, classification-bias study, Whytock et al.

Development order and evidence required

The phases below are a proposed sequence, not calendar commitments or implemented features.

Priority	Concrete change	Evidence required before adoption
P0	Expert-reviewed benchmark with stable media/object IDs and site/encounter holdouts	Label audit; no related encounters crossing splits; all failures and unknowns retained
P0	Confusion/error dashboard and stratified audit queue	Reproducible per-class, day/night and end-to-end metrics with group-based uncertainty
P0	Operational-effort, clock and data-quality audit	Correct active/inactive histories and traceable edits on representative projects
P1	Camtrap DP and camtrapR-compatible exports	Schema validation and agreement with a pinned reference workflow
P1	Local model comparison and optional SpeciesNet fine-tuning	Matched field holdouts; learning curves; no unexplained regression in priority taxa

Priority	Concrete change	Evidence required before adoption
P1	Calibration and review policy	Independent calibration subset; reliability and risk/coverage evaluation; rare-class limitations shown
P1	Activity sensitivity, overlap and coverage-based diversity	Numerical agreement with reference methods and reported assumptions/uncertainty
P2	Single-species occupancy, then community models	Sampling assumptions checked; fit/convergence diagnostics; comparison with expert-derived results
P2	Targeted detector/crop improvements, augmentation and sequence context	Paired ablations on the same holdout; runtime and review burden included
P3	Individual-ID, density models, aerial imagery or audio	A separate validated use case, suitable survey inputs and component licence review

For a first model experiment, fit learning curves at feasible per-class sample levels rather than collecting a large number of nearly identical burst frames. Use the actual target Windows hardware for throughput and cancellation/recovery checks. Preserve a known-working model and activate a candidate only after its evaluation is recorded.

OCR, GBIF matching and Qwen assistance can remain separate document/taxonomy work alongside this sequence. Their outputs should retain source evidence and human-review status.

Current community and software updates

Ten dated community items and eleven technical references are catalogued separately in COMMUNITY_UPDATES.md. The highest immediate relevance is assisted annotation, recoverable video intake, review-interface design and Windows reliability. The October drone/audio announcements are useful watch items for future use cases. Community performance numbers are author reports, not validation of Wildlitics.

The research supports a disciplined route to a better application: trusted survey data, measured local identification, reproducible ecological analysis, and upgrades accepted on evidence. Production integration still depends on the actual application source and representative labelled field data.

Priority evidence and its limits

28 sources assessed through abstracts, accessible article sections or author/publisher method content. These notes distinguish reported findings from proposed Wildlitics changes. No performance number is a local Wildlitics result.

1. Local SpeciesNet fine-tuning

AI and computer vision for wildlife identification in camera trap images: Fine-tuning SpeciesNet outperforms local models for species classification. Science of The Total Environment, 2026. DOI

Reported evidence: A comparison using 15 common Australian rainforest species favoured local fine-tuning of SpeciesNet, including evaluation on withheld cameras.

Limit: The balanced common-species experiment does not establish results for rare Bornean taxa. Its reported learning-curve plateau is not a universal sample requirement.

Wildlitics proposal: Compare a frozen local head and modest SpeciesNet fine-tuning on identical site and encounter holdouts before selecting a backbone.

2. Crop-based classification

To crop or not to crop: Comparing whole-image and cropped classification on a large dataset of camera trap images. IET Computer Vision, 2024. DOI

Reported evidence: Across three datasets, detector-based crops improved classification relative to whole-image baselines.

Limit: Missed detections remain a bottleneck; the paper did not evaluate multi-species images or prove gains for every size of animal.

Wildlitics proposal: Retain crop-to-object associations and benchmark missed animals alongside species classification.

3. Habitat and camera generalisation

Can CNN-based species classification generalise across variation in habitat within a camera trap survey?. Methods in Ecology and Evolution, 2023. DOI

Reported evidence: Classification generalisation deteriorated at unseen cameras and habitat disturbance levels; crops and more varied training backgrounds helped in the study.

Limit: Higher random-split accuracy did not guarantee performance in a new habitat.

Wildlitics proposal: Hold out complete sites/deployments and test across habitat, camera hardware, day/night and seasons.

4. Data volume and background shift

Three critical factors affecting automated image species recognition performance for camera traps. Ecology and Evolution, 2020. DOI

Reported evidence: Performance varied with class training volume and unseen locations in a modest-sized, imbalanced camera-trap collection.

Limit: Reported image counts and accuracy thresholds are dataset-specific.

Wildlitics proposal: Publish per-species learning curves and use encounter counts as well as image counts when allocating labelling effort.

5. Active learning

A deep active learning system for species identification and counting in camera trap images. Methods in Ecology and Evolution, 2021. DOI

Reported evidence: Combining human annotation, transfer learning and object crops reduced annotation requirements in the studied workflow.

Limit: Its large annotation-saving estimate is not a forecast for Wildlitics.

Wildlitics proposal: Offer a diverse review queue covering model disagreement, uncertain cases, unfamiliar backgrounds and rare classes.

6. Confidence calibration

Being confident in confidence scores: calibration in deep learning models for camera trap image sequences. Remote Sensing in Ecology and Conservation, 2025. DOI

Reported evidence: A European camera-trap study examined calibration over five models and independent test sets; sequence-logit averaging and optional temperature scaling were useful.

Limit: Calibration can change under domain shift. Sequences can contain multiple animals or species.

Wildlitics proposal: Use a separate calibration subset, retain logits when available and evaluate Brier score, log loss and reliability plots.

7. Context across animal crops

Paying attention to other animal detections improves camera trap classification models. Methods in Ecology and Evolution, 2026. DOI

Reported evidence: A learned attention module used context from crop embeddings within images/sequences to improve classification over independent-crop baselines.

Limit: Part of the multi-species evaluation used synthetic or weakly annotated sequences.

Wildlitics proposal: Explore context after a baseline is measured; preserve individual object predictions instead of forcing one species across an encounter.

8. Background-diverse augmentation

Copy-Paste Augmentation Improves Automatic Species Identification in Camera Trap Images. Ecology and Evolution, 2025. DOI

Reported evidence: Copy-paste training augmentation improved generalisation to unseen locations in the reported experiments.

Limit: Benefits varied by species, and very small training sets produced mixed results.

Wildlitics proposal: Test augmentation only after group splitting, with traceable source masks and no held-out backgrounds entering training.

9. Training quality and ecological error

Deep learning-based ecological analysis of camera trap images is impacted by training data quality and quantity. Remote Sensing in Ecology and Conservation, 2026. DOI

Reported evidence: Rare or visually similar species were especially sensitive to training quality/quantity; species-level occupancy and activity could diverge even when community richness appeared stable.

Limit: The study used two ecosystems and does not prove that architecture choice never matters.

Wildlitics proposal: Prioritise clean labels and class coverage, then evaluate ecological outputs against expert-labelled data.

10. AI-derived occupancy: promising but conditional

Identification of camera trap images by artificial intelligence and human experts produces similar multi-species occupancy models. Journal of Applied Ecology, 2026. DOI

Reported evidence: After specified processing, AI- and expert-derived multi-species occupancy results were often similar across three study areas.

Limit: Rare species showed some of the largest divergences; agreement depended on processing choices.

Wildlitics proposal: Build an ecological agreement benchmark before allowing unattended AI-labelled analyses.

11. Validate the ecological endpoint

Robust ecological analysis of camera trap data labelled by a machine learning model. Methods in Ecology and Evolution, 2021. DOI

Reported evidence: A Central African study directly compared richness, activity and occupancy from AI and expert labels; acceptance thresholds affected the endpoints differently.

Limit: Agreement for several tested species does not establish general equivalence or one best threshold for every analysis.

Wildlitics proposal: Compare image errors and downstream ecological estimates separately, documenting rejected and unresolved records.

12. Precision, night images and occupancy bias

Essential tools but overlooked bias: Artificial intelligence and citizen science classification affect camera trap data. Methods in Ecology and Evolution, 2025. DOI

Reported evidence: An expert holdout and simulations showed classification error varying by species and illumination; low precision could substantially bias occupancy estimates.

Limit: The compared classifiers and species groups do not cover all camera-trap deployments.

Wildlitics proposal: Report day/night per-class precision and recall, and separately audit rare-species false positives.

13. Formal data quality control

A protocol for error prevention and quality control in camera trap datasets. Journal of Applied Ecology, 2025. DOI

Reported evidence: The proposed protocol covers malfunction checks, time/storage corrections, classification and post-classification quality control.

Limit: A protocol reduces avoidable errors but cannot establish that every identification or timestamp is correct.

Wildlitics proposal: Add a pre-analysis quality report covering operational effort, clock edits, processing failures, duplicate media and reviewer audit samples.

14. Evidence against routine independence filters

The inappropriate use of time-to-independence biases estimates of activity patterns of free-ranging mammals derived from camera traps. Ecology and Evolution, 2022. DOI

Reported evidence: The study found that time-to-independence filtering altered activity estimates and discarded substantial information, with differing effects across taxa.

Limit: Its conclusion is debated; the contrasting Smith study below identifies behavioural reasons filters may sometimes help.

Wildlitics proposal: Retain raw detections and make filtered/unfiltered activity results available with their sampling units.

15. Counter-evidence on independence filters

An empirical assessment of the role of independence filters in temporal activity analyses using camera trapping data. Behavioral Ecology and Sociobiology, 2025. DOI

Reported evidence: An empirical counter-argument found that lingering or resting animals and herding could distort activity densities, and that filtering could address those behaviours.

Limit: This is not proof that a fixed 60-minute rule is suitable for every habitat or estimator.

Wildlitics proposal: Offer sensitivity comparisons and record the biological rationale for the chosen activity treatment.

16. Occupancy and temporal autocorrelation

Guidelines for estimating occupancy from autocorrelated camera trap detections. Methods in Ecology and Evolution, 2024. DOI

Reported evidence: Simulations and a mammal case study evaluated autocorrelation; larger detection windows could mitigate bias, while inserting gaps discarded data without solving it.

Limit: Appropriate occasion duration depends on detection rates, design and model assumptions.

Wildlitics proposal: Build explicit occasion histories, test fit/autocorrelation and distinguish occasion width from encounter grouping.

17. Interoperable camera-trap records

Camtrap DP: an open standard for the FAIR exchange and archiving of camera trap data. Remote Sensing in Ecology and Conservation, 2024. DOI

Reported evidence: Camtrap DP specifies linked deployment, media and observation tables supporting human and AI classification workflows.

Limit: A valid exchange package does not validate survey design or scientific interpretation.

Wildlitics proposal: Add a version-pinned Camtrap DP export with schema validation and stable IDs while preserving raw AI and accepted observations.

18. camtrapR foundations

camtrapR: an R package for efficient camera trap data management. Methods in Ecology and Evolution, 2016. DOI

Reported evidence: The package links camera-trap organisation, extraction and summaries to detection histories for occupancy and capture-recapture analysis.

Limit: Species and individual identification require their own workflows and quality control.

Wildlitics proposal: Export compatible record tables and camera-operation matrices rather than inventing a separate ecological data format.

19. camtrapR version 3: a current integration reference

The camtrapR R package: From data management to interactive ecological analysis of camera trap data. Methods in Ecology and Evolution, 2026. DOI

Reported evidence: Published online on 21 October 2026, the paper describes an end-to-end workflow including a dashboard, covariates, occupancy models, spatial predictions and diagnostics.

Limit: Advanced models still require appropriate designs, covariates and diagnostic interpretation; a dashboard does not remove those requirements.

Wildlitics proposal: Evaluate a local R analysis adapter and reusable exports before independently reimplementing every statistical method.

20. Occupancy with imperfect detection

ESTIMATING SITE OCCUPANCY RATES WHEN DETECTION PROBABILITIES ARE LESS THAN ONE. Ecology, 2002. DOI

Reported evidence: The occupancy framework distinguishes species occurrence from the probability of observing it during repeat surveys.

Limit: Closure, sampling and detection assumptions remain necessary; an unprocessed or inactive camera occasion is not a nondetection.

Wildlitics proposal: Export 1/0/NA histories based on observed detections and documented operational effort, with model assumptions visible.

21. False-positive occupancy

GENERALIZED SITE OCCUPANCY MODELS ALLOWING FOR FALSE POSITIVE AND FALSE NEGATIVE ERRORS. Ecology, 2006. DOI

Reported evidence: The model extends occupancy analysis to include false-positive as well as false-negative observations.

Limit: Such models need an identifiable design and suitable validation information; AI scores alone are not automatically enough.

Wildlitics proposal: Preserve validation/audit observations so future error-aware models have defensible inputs.

22. Daily activity overlap

Estimating overlap of daily activity patterns from camera trap data. Journal of Agricultural, Biological, and Environmental Statistics, 2009. DOI

Reported evidence: The paper compares methods for estimating overlap between circular daily activity distributions.

Limit: Overlap describes temporal coincidence and does not establish species interaction or causation.

Wildlitics proposal: Add tested circular overlap estimates and uncertainty, reporting estimator, sample sizes and preprocessing.

23. Activity level needs extra assumptions

Quantifying levels of animal activity using camera trap data. Methods in Ecology and Evolution, 2014. DOI

Reported evidence: The proposed activity-level estimator uses time-of-detection distributions and an assumption about activity at the daily peak.

Limit: The proportion of time active cannot be read directly from a normalised histogram or a basic density curve.

Wildlitics proposal: Keep descriptive timing charts distinct from an optional, assumption-documented activity-level model.

24. Diversity with sample coverage

iNEXT: an R package for rarefaction and extrapolation of species diversity (H ill numbers). Methods in Ecology and Evolution, 2016. DOI

Reported evidence: iNEXT supports abundance- and incidence-based rarefaction/extrapolation for Hill-number diversity and sample completeness.

Limit: Camera detections are not automatically independent counts of individual abundance.

Wildlitics proposal: Prioritise incidence-based sampling units, coverage curves and uncertainty for camera-trap diversity comparisons.

25. Comparable sampling completeness

Coverage-based rarefaction and extrapolation: standardizing samples by completeness rather than size. Ecology, 2012. DOI

Reported evidence: Coverage-based rarefaction/extrapolation compares communities at similar sample completeness instead of merely equal sample size.

Limit: Appropriate sampling units and uncertainty still matter.

Wildlitics proposal: Display effort and completeness next to richness comparisons, with clearly stated incidence units.

26. Individual identification is a separate error problem

Identification errors in camera-trap studies result in systematic population overestimation. Scientific Reports, 2020. DOI

Reported evidence: A controlled snow-leopard identification experiment showed that human individual-ID errors could inflate abundance estimates.

Limit: This concerns individual recognition and capture-recapture, not a measured error rate for species classification in Wildlitics.

Wildlitics proposal: Keep future individual-ID features separate and require their own independent validation before density estimation.

27. Unmarked density: required inputs

Estimating animal density using camera traps without the need for individual recognition. Journal of Applied Ecology, 2008. DOI

Reported evidence: The random encounter model relates encounter rates to density using movement, group size and camera detection geometry.

Limit: Camera placement and parameter measurement affect validity; encounter counts alone do not identify population density.

Wildlitics proposal: Defer a density button until the survey and calibration inputs required by its chosen estimator are available.

28. CameraSweet's methodological roots

Automatic Storage and Analysis of Camera Trap Data. The Bulletin of the Ecological Society of America, 2010. DOI

Reported evidence: The article presents structured storage and automated tabulation to reduce manual data entry and make camera-trap analyses easier to combine.

Limit: Historical directory-based conventions need adaptation to modern per-object AI records and explicit observation provenance.

Wildlitics proposal: Adopt reproducible tables, clock correction history and documented effort while preserving original media.